Rheology

Protocol for the laboratory course in Polymer chemistry

Universität of Stuttgart

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Authors: Giovanni Costa (3651964)
st184329@stud.uni-stuttgart.de

Nicolas Czimber (3679894)
st185659@stud.uni-stuttgart.de

Assistant: Semi Kim

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Contents

1 Introduction and theory [1] 1

1.1 Viscoelastic behavior and oscillation measurement 3

2 Task 4

2.1 Investigation of flow behavior 4

2.2 Time dependent rheological behavior 4

2.3 Oscillation measurements 4

3 Experimental procedure 4

3.1 Investigation of flow behavior 4

3.2 Time dependent rheological behavior 4

3.3 Oscillation measurements 5

4 Analysis 6

4.1 Investigation of flow behavior 6

4.2 Time dependent rheological behavior 8

4.3 Oscillation measurements 8

4.3.1 Amplitude test 8

4.3.2 Frequency test 10

5 Discussion 12

6 Questions 12

7 References 13

1 Introduction and theory [1]

Rheology deals with the flow and deformation behavior of liquids and solids. In general there are two extremes of rheological behavior. On one side there is elastic behavior, on which a constant shear stress leads to a constant deformation, which is completely reversible. This behavior is characteristic for solids. On the other side there is viscose behavior, where a linear, irreversible correlation between shear stress and deformation is observable. This behavior is characteristic for liquids. Not all substances just feature elastic or viscose behavior, but a combination of both depending on the kind of the applied deformation. This behavior is called viscoelastic. So viscoelasticity describes the property of a substance showing both fluid- and solid-like properties, depending on the applied deformation.
One model for illustrating rheological behavior in detail is the two-plate model. It is schemed in figure 1.

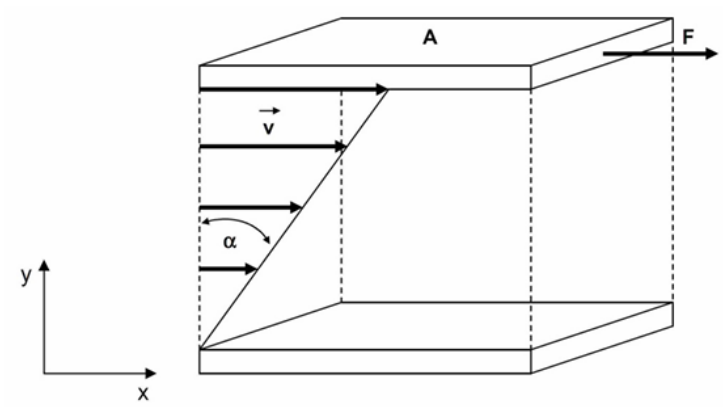

Figure 1: Schematic image of the two-plate model. [1]

In this model a liquid, which is between two parallel plates is viewed. An upper plate, with an area of A is now moved with a force F and is inducing a shearing deformation γ. The velocity of this deformation $\dot{\gamma}$, also called shear rate, is dependent on the distance to the moving plate. Moreover, the shear stress τ and viscosity η are two important dimensions. They are defined as follows.

$$\tau = \frac{F}{A} = \eta \cdot \dot{\gamma} \tag{1.1}$$

$$\eta = \frac{\tau}{\dot{\gamma}} \tag{1.2}$$

The correlation between shear stress, viscosity and deformation velocity is important to understand the structure of a material. In figure 2 the dependence of the shear stress and the deformation velocity of three different types of liquids is shown.

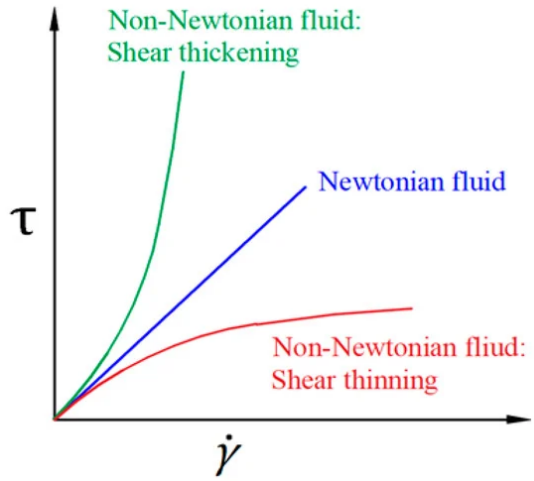

Figure 2: Correlation between shear stress and shear rate of a newtonian, a shear thickening and a shear thinning liquid. [2]

There are newtonian fluids. They are showing a linear correlation between shear stress and shear rate. There are also non-newtonian liquids. On one side are shear thickening liquids. They feature an exponentially increasing shear stress with higher deformation velocity. On the other side are shear thinning liquids. They feature a limited exponential progression of the shear stress.
The correlation of viscosity and shear rate of these kinds of liquids is shown in figure 3.

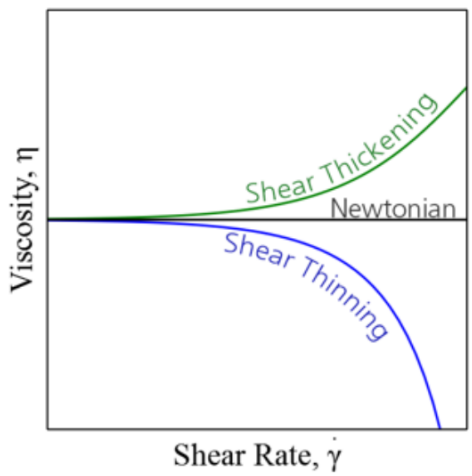

Figure 3: Correlation between the viscosity and shear rate of a newtonian, a shear thickening and a shear thinning liquid. [3]

In newtonian liquids the viscosity is independent of the shear rate. In shear thickening liquids the viscosity increases with higher shearing rates, while in shear thinning liquids the viscosity decreases.

Another important aspect is the time dependence of the viscosity of liquids. In thixotropic liquids the liquid experiences a dismantling of structure and complete structure build-up

after a certain time is waited after the relief. Non-thixotropic do not feature a structure build-up, or the structure build-up after a deformation is incomplete. The last category are rheopexic liquids, where the build-up of the structure is happening under a constant deformation. In relief a dismantling is observed.

1.1 Viscoelastic behavior and oscillation measurement

The viscoelastic behavior can be analysed in oscillation measurementS were an oscillating deformation is applied, leading to a shear stress oscillating out of phase, with a phase shift δ. This viewing is just valid for the linear-viscoelastic range of deformation, in which the elastic and viscose part of the rheological behavior is considered separable. Regarding viscoelastic behavior there are two dimensions, which describe the elastic and the viscose share of the deformation. It is the storage module G' for the elastic share and the loss module G'' for the viscose share. The storage module is considered the part of the energy, which is getting reversibly stored in the liquids structure, when a deformation is applied, while the loss module is considered as the part of the energy, which is getting irreversibly stored, when the deformation is applied.
There is a connection between the storage module, loss module and the phase shift in the oscillation measurement. It is shown in the following equation.

$$\tan\delta = \frac{G''}{G'} \tag{1.3}$$

$\tan\delta$ is called the loss factor, or the share of energy, which is irreversibly stored. It is lower than 1 for liquids where the elastic behavior is dominant and bigger than 1 for liquids where the viscose behavior is dominant. For pure elastic behavior the loss factor is equal to 0, leading to no observed phase shift in the oscillation measurement. For pure viscose behavior the loss factor tends to infinity, leading to an observed phase shift of $\frac{\pi}{2}$. With these definitions it is possible to define the value of the complex viscosity $|\eta^*|$.

$$|\eta^*| = \frac{\tau_0}{\dot{\gamma}_0} \tag{1.4}$$

In the oscillation measurement the viscosity is the ratio between the shear stress and the shear rate.

To conduct an oscillation measurement, several kinds of geometries are possible. One, already mentioned, geometry is the plate-plate geometry. Another one is plate-cone geometry. In there the upper, moving plate is substituted to a rotating cone. This brings the advantage of the independence of the shear stress from the distance to the center of the cone.
In this geometry rotational measurements are also possible to conduct. In there the cone is rotating with a specific velocity. The oscillation measurement itself can be conducted using a constant frequency, while varying the amplitude (amplitude test), or by varying the frequency at a constant amplitude (frequency test). It is practical to first conduct the amplitude test for investigating the range of the linear viscoelastic behavior and setting an amplitude for the frequency test, so that the measurement is conducted in the range, where linear viscoelastic behavior is observed.

2 Task

2.1 Investigation of flow behavior

Different liquids should be classified in their characteristic flow behavior. Moreover, this behavior is discussed regarding the liquids structure. The investigated liquids are honey, a PVP solution of 14 wt% in water and a starch suspension in water.

2.2 Time dependent rheological behavior

The time dependence of the viscosity of Ketchup is analysed. Moreover, this behavior is traced back to the structure.

2.3 Oscillation measurements

On a PVP solution of 20 wt% in water, the viscoelastic behavior is investigated by using oscillation measurements. Thereby the characteristic change in the flow behavior should be illustrated.

3 Experimental procedure

All measurements were conducted at 25 °C in a plate-cone geometry. The angle of the cone was 1.004°.

3.1 Investigation of flow behavior

For the investigation of the flow behavior, rotational measurements of the samples were done. Shear stress and viscosity were measured.
In the measurement the deformation velocity $\dot{\gamma}$ was varied. For the honey sample it was varied, beginning from $0.1\,\mathrm{s}^{-1}$ and ending with $186\,\mathrm{s}^{-1}$. For the PVP solution ($M_w = 1\,300\,000\,\mathrm{g/mol}$) of 14 wt% the velocity started at $5000\,\mathrm{s}^{-1}$ and ended at $1\,\mathrm{s}^{-1}$. The starch suspension in water had a weight share of 50 % of starch and was measured at velocities of deformation between $1\,\mathrm{s}^{-1}$ and $100\,\mathrm{s}^{-1}$.

3.2 Time dependent rheological behavior

To investigate the time dependency of the change in the viscosity of a ketchup sample the following measurement program was conducted. For the first twenty seconds the sample was deformed with a velocity of $0.25\,\mathrm{s}^{-1}$. In the following ten seconds the velocity is reduced to $0.1\,\mathrm{s}^{-1}$ just before increasing it to $500\,\mathrm{s}^{-1}$ and holding it for another ten seconds. After that the deformation velocity is again reduced to $0.1\,\mathrm{s}^{-1}$, holding that velocity for 125 s. This program was then repeated twice.

3.3 Oscillation measurements

For the oscillation measurement first an amplitude test was done with the PVP solution ($M_w = 360\,000\,g/mol$) of 20 wt%. Therefore, the deformation amplitude γ_0 was varied from 5000 % to 1 % at a frequency of 10 rad/s. From the amplitude test an amplitude of 70.8 % was selected. This was done ensuring the following frequency test was done in the linear viscoelastic arrange. The frequency in the frequency test was varied from $500\,s^{-1}$ to $1\,s^{-1}$.

4 Analysis

4.1 Investigation of flow behavior

In figure 4 both the flow curve and the viscosity curve of the honey sample are depicted. In there the dependency of the shear stress τ and the deformation velocity $\dot{\gamma}$ as well as the dependency of the viscosity η and the shear rate are shown.

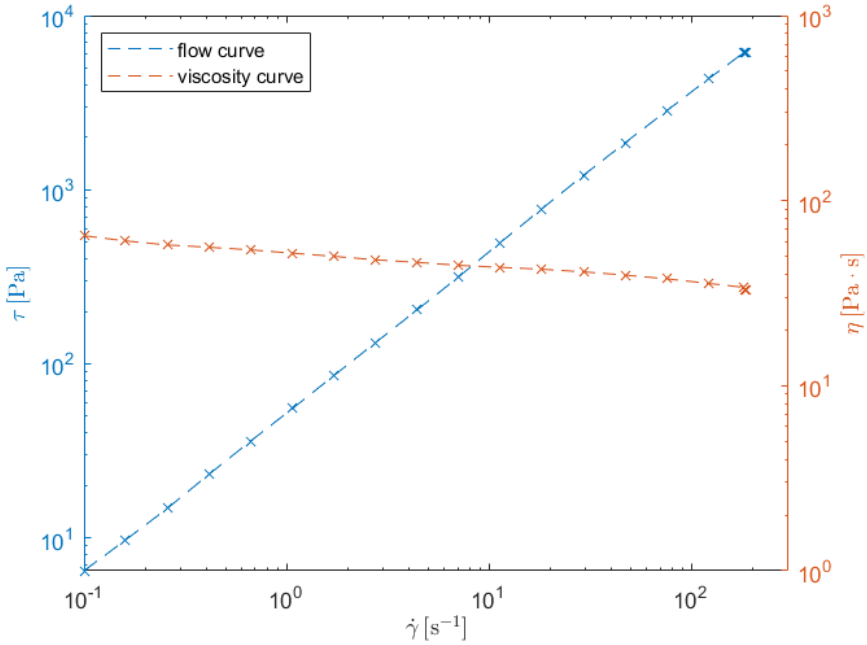

Figure 4: Flow curve and viscosity curve of the honey sample.

It is noticeable that, both the graphs of the flow and the viscosity curves shows the characteristics of a newtonian liquid. While the viscosity stays on an almost constant value, when increasing the deformation velocity, the shear stress exhibits a linear increase. The small decrease in the viscosity is due to inhomogeneities, like crystallized sugar, or air bubbles, disturbing the ideal flow behavior of the honey.

In figure 5 the flow and viscosity curve of the PVP solution of 14 wt% is shown.

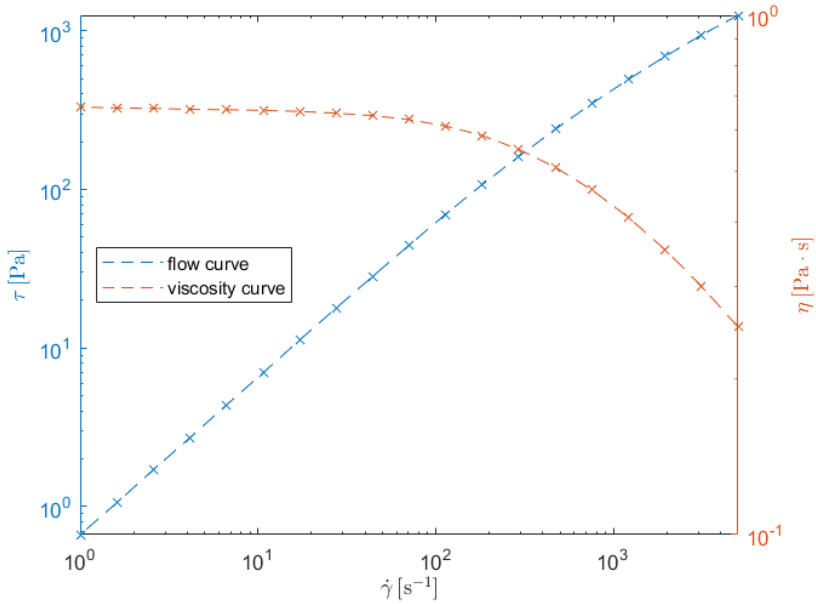

Figure 5: Flow curve and viscosity curve of the PVP solution.

The polymer solution shows a shear thinning behavior, with the viscosity drastically decreasing at higher deformation velocity. This means the solution gets less viscose, when more pressure is applied. This is because of orientation of the polymer chains in a solution. At low applied deformation velocity the polymer chains tangle together due to interactions. But if the deformation gets faster the polymer chains orientate in the direction of shearing. If the chains are orientated they can move more easily. This results in a lower viscosity.

In figure 6 the flow and viscosity curve of the starch suspension is visible.

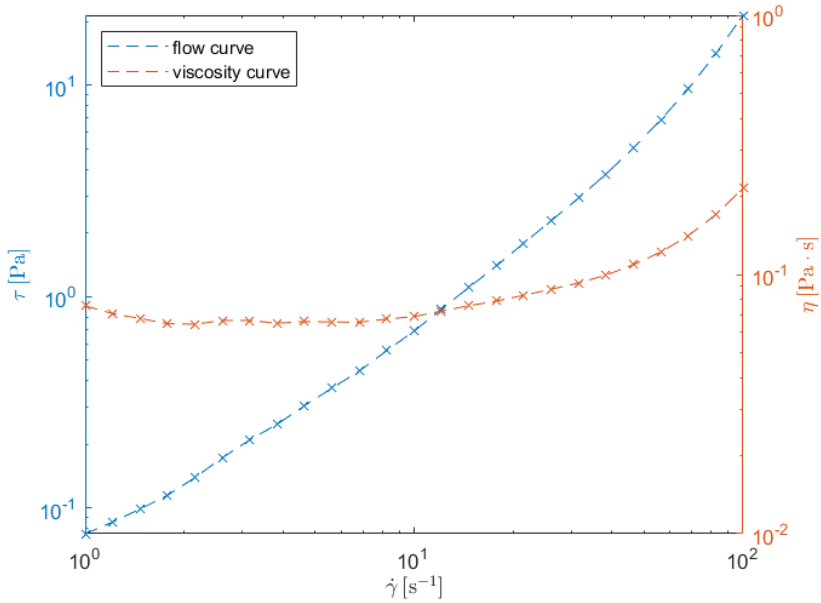

Figure 6: Flow curve and viscosity curve of the starch suspension.

The starch suspension behaves as a shear thickening liquid, with the viscosity drastically increasing by applying a higher deformation velocity. This is because at higher velocity the less viscose water molecules are more separated from the starch chains, leading to more contact between them. This results in a higher viscosity.

4.2 Time dependent rheological behavior

Figure 7 shows the viscosity of ketchup, when repeating the same deformation program for three times, with no pause in between.

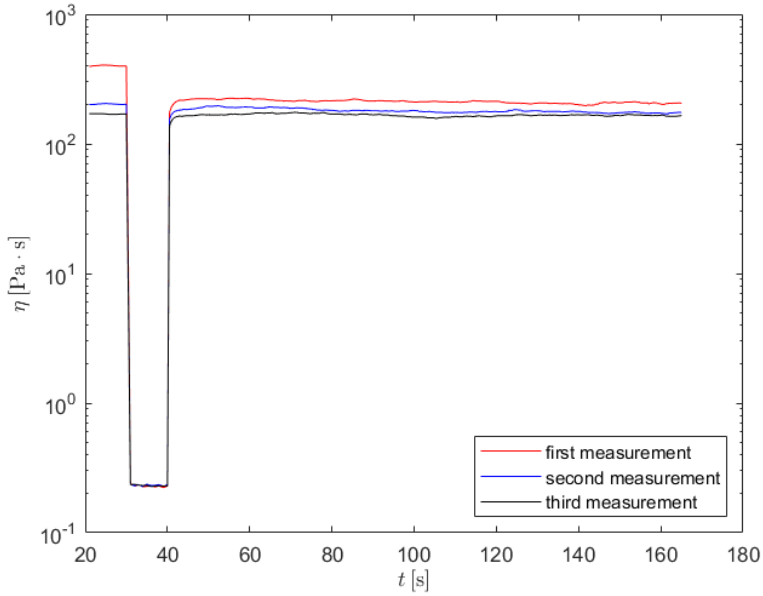

Figure 7: Viscosity of ketchup in dependence of the progress of the rheological measurement.

It is noticeable, that after applying a high deformation velocity for several seconds and returning to the same deformation velocity than before, the ketchup than shows a lower viscosity. Moreover, in the next measurements the same effect is observable. Even though not that drastically than in the first measurement. This behavior is typical for thixotropic liquids like ketchup. In ketchup polymers are added to thicken the product. In solution these polymers show a shear thinning behavior, like the PVP solution. If the applied force is now reduced the orientated polymer chains do not tangle together instantly but with time. This leads to the viscosity being lower than before. This explains the time dependency of this thixotropic liquid.

4.3 Oscillation measurements

4.3.1 Amplitude test

In figure 8 and 9 the correlation between the storage module G' and the loss module G'' of the PVP solution of 20 wt% with the deformation caused by different amplitudes in the

oscillation measurement is shown.

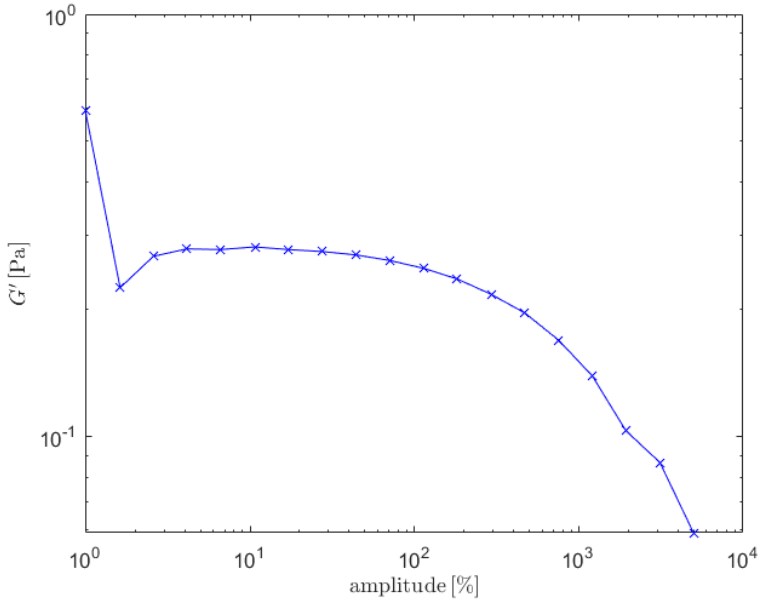

Figure 8: Correlation between storage module of the PVP solution and its amplitude in deformation.

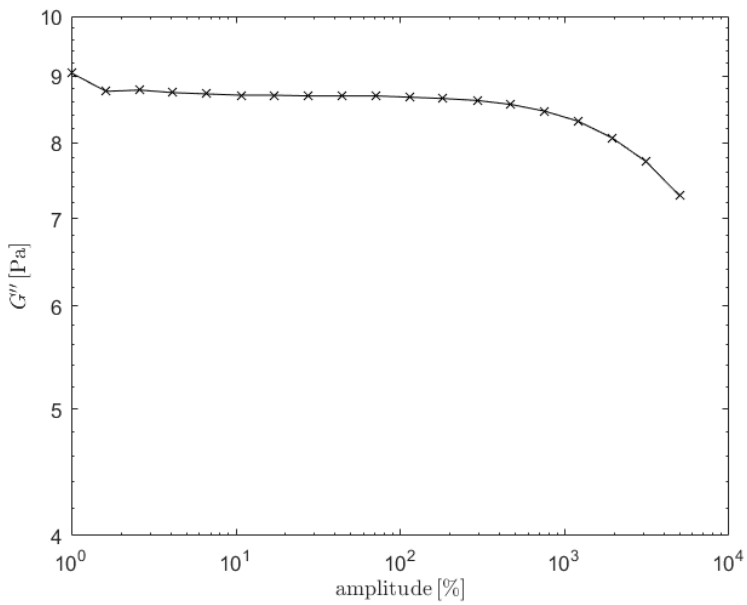

Figure 9: Correlation between loss module of the PVP solution and its amplitude in deformation.

In the following figure 10 the absolute value of the complex viscosity in dependence of the amplitude is depicted.

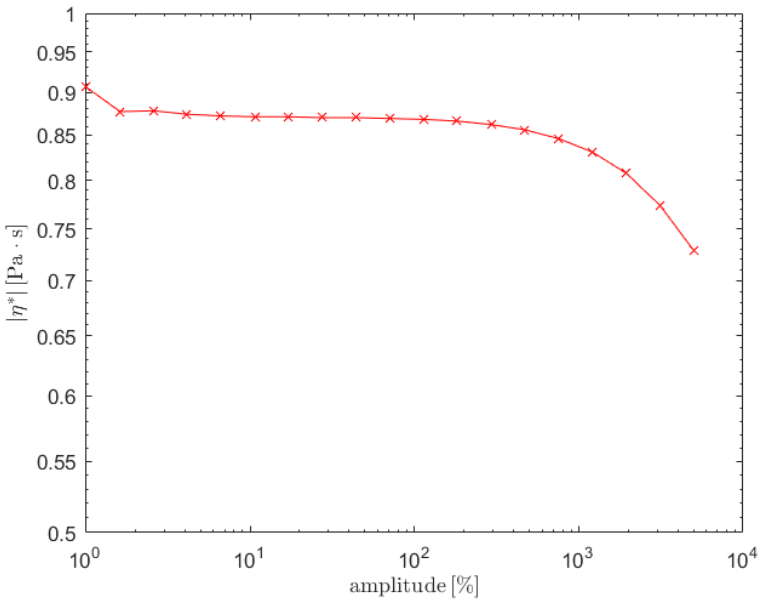

Figure 10: Correlation between value of the complex viscosity of the PVP solution and its amplitude in deformation.

It is noticeable, that with higher deformation the storage and loss module first stays nearly constant and then later drastically decreases. The same course of the graph is observed with the viscosity. Hereby the loss module decrease more than the storage module. The range where both the modules are constant is called the linear viscoelastic range. This range is characterized by the property, that the storage module and the loss module can be viewed separately. This linear viscoelastic behavior is observable up to a deformation of about 75 %. Because of that an amplitude of 70.8 % was chosen for the following frequency test.

4.3.2 Frequency test

In figure 11 the dependence of the deformation frequency and both the storage and the loss module is illustrated.

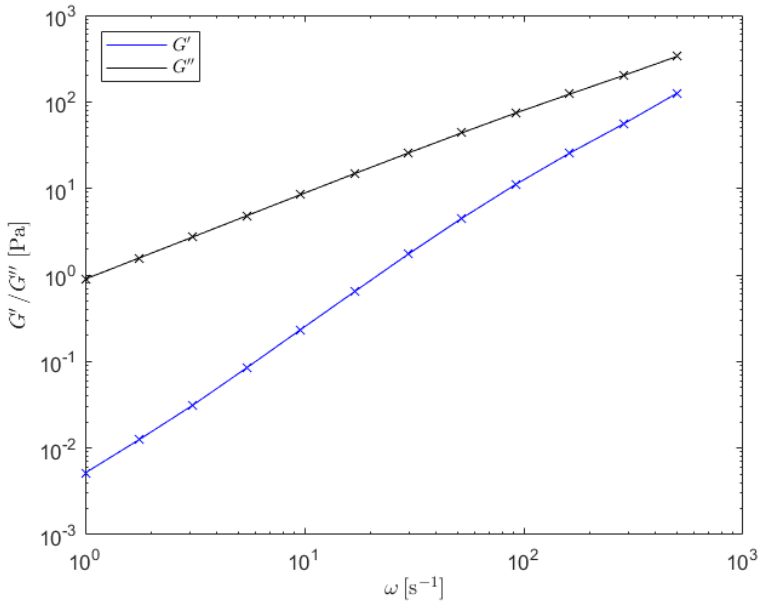

Figure 11: Correlation between frequency, storage and loss module.

The dependence of the value of the complex viscosity and the frequency is shown in figure 12.

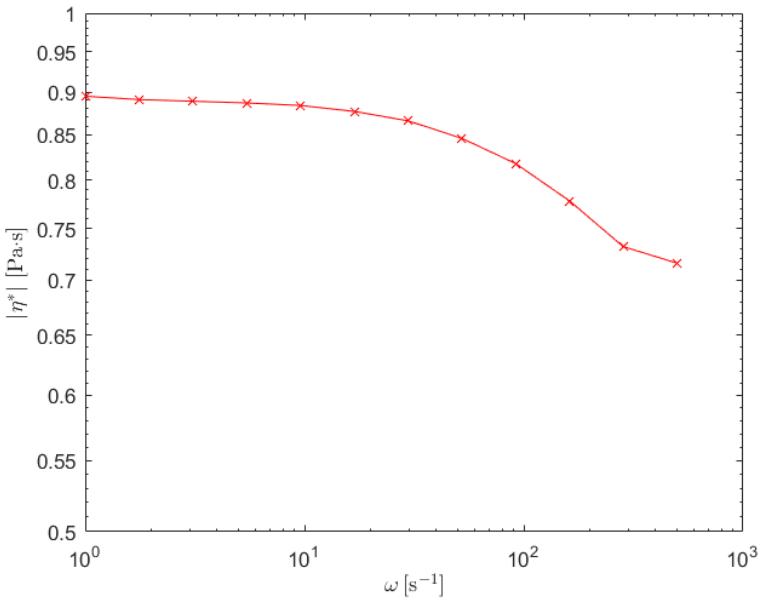

Figure 12: Correlation between complex viscosity and frequency.

It is shown, that with higher frequency both the storage and the loss module gets higher. However, as mentioned earlier, the storage module increases more than the loss module, leading to the loss factor getting smaller. This means, that with higher frequencies the elastic behavior gets more dominant, while at lower frequencies the viscose behavior is dominant.

The reason of this lies in the shear thinning behavior of the PVP solution. With higher frequency the polymer chains get more orientated in the direction of the deformation. Therefore, the chains get less flexible, which allows them to reversibly store more and more deformation energy. This leads to a lower loss factor and an increase of the elastic share of the liquid, compared to the viscose one. This is also observable in the drastic decrease of the viscosity, depicted in figure 12.

5 Discussion

Like all experimental methods, also the method used in this work are influenced by inaccuracies. The biggest source for these inaccuracies might be impurities and contamination. The influence of impurities can be seen in the viscosity curve in figure 4, where gas bubbles and crystallized sugar leads to deviations from the newtonian behavior. Another aspect might be contamination. After every measurement the plate and the cone had to be cleaned properly. So some leftover sample from past measurements could have influenced the gained values.

6 Questions

1. Enlist examples of methods, which can be applied to determine the viscosity of a material.

To determine the viscosity of a liquid, it is also possible to analyse the flow behavior using viscometry. In it the time is measured in which a liquid is wandering a specific distance due to gravity. Another method is the ball drop method. This method is measuring the time it takes a ball to wander through a liquid at rest.

2. Outline the most important flow and viscosity diagrams.

The most important flow and viscosity diagrams are already depicted and described. They can be found in figure 2 and 3.

3. Explain the advantages and disadvantages of the plate-plate and the plate-cone geometry.

In the plate-plate geometry it is possible to select a higher distance between the two plates, than it would be possible for the distance between the plate and a cone. Therefore, it is possible to minimize the influence of thermal expansion and other unwanted occurrence of friction. The disadvantage of this geometry is, that in contrast to the plate-cone geometry, the deformation velocity is dependent on the distance to the center of moving plate.

7 References

[1] *Script to "laboratory course in polymer chemistry"*, University of Stuttgart, **2025**.

[2] Y. Song, Z. Zhuang, X. Wang, Q. Fang, Z. Cheng, T. Zhang, *Actuators* **2024**, *13*.

[3] G. Baek, Great Customer Question: Can You Measure Zero Shear Viscosity, `https://blog.rheosense.com/great-customer-question-can-you-measure-zero-shear-viscosity`, **2020**.